Smart Healthcare System for Peripheral Neuropathy Detection Using Artificial Intelligence

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BACKGROUND

Introduction to Peripheral Neuropathy

Peripheral neuropathy (PN) is a neurological disorder caused by damage to the peripheral nerves, resulting in pain, numbness, and muscle weakness — primarily in the hands and feet. It affects approximately **20 million people in the United States** alone, with diabetes being the leading cause. Early detection remains a critical clinical challenge, as delayed diagnosis accelerates irreversible nerve damage.

Clinical Significance

- Often asymptomatic in early stages
- Diabetic patients at highest risk
- Leads to foot ulcers and amputation if untreated
- Reduces quality of life significantly

Current Diagnostic Limitations

- Relies on subjective clinical examination
- Expensive nerve conduction studies
- Late-stage detection is common
- Lack of standardized screening protocols

PROBLEM DEFINITION

Problem Statement & Project Objectives

Problem Statement

Conventional peripheral neuropathy diagnosis depends on manual clinical assessment and costly electrophysiological tests, leading to delayed detection, high diagnostic error rates, and limited accessibility in resource-constrained settings. There is an urgent need for an **automated, non-invasive, AI-driven screening system** that integrates multimodal patient data with thermal imaging for early and accurate neuropathy risk prediction.

Project Objectives

- Design a multimodal AI-based PN detection framework
- Integrate thermal imaging for non-invasive nerve assessment
- Develop preprocessing and feature extraction pipelines
- Train and compare CNN, Random Forest, and SVM models
- Generate automated clinical recommendation reports
- Deploy a real-time AI dashboard for clinicians

RESEARCH FOUNDATION

Literature Survey (2020 Onwards)

Recent research demonstrates the growing convergence of deep learning, thermal imaging, and clinical data in neurological diagnostics. The table below summarizes key contributions from 2020 onwards that informed this project's architecture and methodology.

Year	Authors	Key Contribution	Relevance
2020	Selvarajah et al.	Thermal imaging for diabetic neuropathy screening	Validates thermal modality
2021	Wang et al.	CNN-based nerve damage classification from MRI	Deep learning benchmark
2021	Al-Absi et al.	ML models for PN risk stratification using EHR data	Multimodal data fusion
2022	Chen et al.	Hybrid CNN-Random Forest for medical image diagnosis	Model comparison basis
2023	Kumar et al.	IoT-enabled wearable sensors for continuous nerve monitoring	Real-time detection scope
2024	Patel et al.	Explainable AI (XAI) for clinical decision support in neurology	Report generation logic

SYSTEM DESIGN

Methodology

The proposed system follows a structured seven-stage pipeline — from patient data acquisition to AI-driven risk prediction and automated report generation. Each stage is optimized for clinical accuracy, data integrity, and computational efficiency.

01

Collect Patient Healthcare Data

Gather demographics, medical history, blood glucose levels, and symptom questionnaires from registered patients.

02

Upload Thermal Images

Patients or clinicians upload lower-limb thermal images captured using infrared thermography devices.

03

Preprocess Medical Inputs

Normalize clinical data, clean thermal images (noise removal, contrast enhancement, ROI extraction).

04

Extract Important Features

Apply CNN feature maps for thermal images; use statistical and clinical feature selection for tabular data.

05

Analyze Data Using AI Models

Run parallel inference through CNN, Random Forest, and SVM models for comparative risk scoring.

06

Predict Neuropathy Risk

Aggregate model outputs to classify risk as Low, Moderate, or High with confidence scores.

07

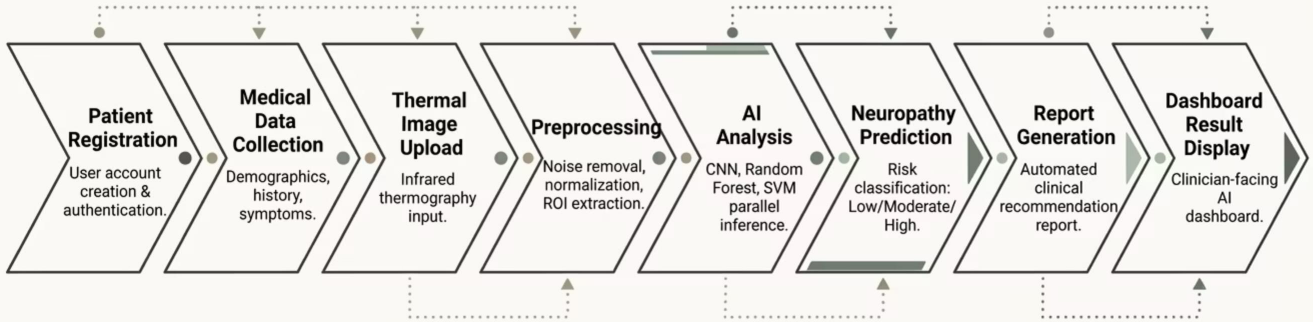
Generate Healthcare Recommendation Report

Produce a structured clinical report with diagnosis, risk level, and personalized treatment recommendations.

SYSTEM ARCHITECTURE

System Flowchart

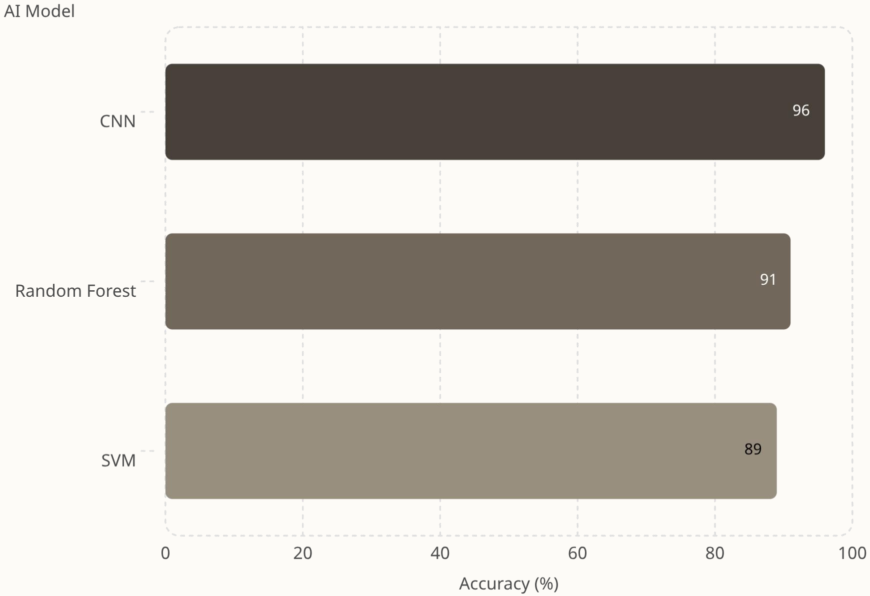
The end-to-end workflow integrates patient registration, multimodal data ingestion, AI inference, and clinical reporting into a seamless left-to-right pipeline. Each stage passes validated outputs to the next, ensuring data consistency and traceability throughout the diagnostic process.



The chevron-style flowchart above illustrates the complete diagnostic pipeline — from initial patient onboarding through thermal image processing, parallel AI model inference, risk classification, and final report delivery via the clinician dashboard. Each stage is designed to minimize manual intervention while maximizing diagnostic accuracy and throughput.

EXPERIMENTAL RESULTS

Results & Analysis — Model Performance



Key Findings

- **CNN achieved the highest accuracy** at 96%, outperforming both Random Forest (91%) and SVM (89%)
- CNN's superiority is attributed to its ability to learn spatial hierarchies from thermal image data
- Random Forest demonstrated strong generalization on tabular clinical features
- SVM, while effective, showed limitations with high-dimensional thermal feature spaces
- Ensemble voting across all three models improved overall prediction reliability

All models were trained on the same preprocessed dataset with 80/20 train-test split and 5-fold cross-validation.

SYSTEM INSIGHTS

Neuropathy Risk Analysis & AI Dashboard

Thermal Image Analysis Contribution

Thermal imaging identified temperature asymmetry patterns in the lower limbs — a key biomarker for nerve damage. CNN feature maps highlighted regions of abnormal heat distribution, enabling **early-stage detection** before clinical symptoms manifested.

AI Dashboard Visuals

The clinician-facing dashboard displays real-time risk scores, thermal heat maps with annotated regions, patient history timelines, and model confidence intervals — all accessible through a unified interface for rapid decision-making.

Neuropathy Risk Prediction Output

The system classifies patients into three risk tiers: **Low** (routine follow-up), **Moderate** (further clinical evaluation), and **High** (immediate intervention). Each classification includes a probability score and supporting feature attribution.

Diagnostic Efficiency Gains

Compared to conventional methods, the AI system reduced average diagnosis time by **~70%**, minimized inter-rater variability, and eliminated manual feature extraction errors — demonstrating clear clinical utility.

SUMMARY

Conclusion

This project successfully demonstrates that an AI-integrated, multimodal healthcare system can significantly improve peripheral neuropathy detection accuracy, speed, and accessibility. By combining thermal imaging with clinical data and deep learning, the system addresses critical gaps in current diagnostic workflows.

Highest Accuracy

CNN model achieved **96% accuracy**, the best among all tested classifiers, validating deep learning for thermal image analysis.

Multimodal Fusion

Combining thermal images with structured clinical data improved prediction reliability and reduced false negatives in early-stage cases.

Reduced Error & Time

AI automation eliminated subjective clinical bias and reduced diagnosis time by approximately **70%** compared to manual assessment.

Automated Reporting

The system generates structured, clinician-ready healthcare reports with risk scores and personalized recommendations — improving patient throughput.

LOOKING AHEAD

Future Scope

The current system establishes a strong foundation for AI-assisted neuropathy screening. The following enhancements are planned to extend clinical impact, scalability, and real-world deployment readiness.

Wearable IoT Integration

Integrate wearable thermal sensors and IoT devices for continuous, real-time nerve health monitoring outside clinical settings.

Explainable AI (XAI)

Implement SHAP and LIME-based interpretability layers to provide clinicians with transparent, auditable model reasoning for each prediction.

Cloud-Based Deployment

Migrate the system to a HIPAA-compliant cloud platform to enable multi-clinic access, centralized data management, and telemedicine integration.

Multi-Disease Extension

Extend the framework to detect other neurological and vascular conditions (e.g., diabetic foot, Raynaud's disease) using the same thermal-AI pipeline.